

• Week 2 ITAI 2372

• Health Care Industry

- Hospitals
- Health Insurance
- Nursing homes
- Pharmacists
- etc

• Use case of AI

- Medical Imaging
- large amount of data
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- Potential Diagnosis
 - AI can review large data sets across long time periods

- Predictive analytics
 - health care mgmt system on a database
 - patient has several markers that flag high risks.
 - Systems that can study these factors.
 - TB is still a risk in India. There are markers that increase this risk.

- System can study lifestyle & profession.
- Systems can give intervention plan
- System can give a personalized treatment plans

● Hospital Admin Tasks

- Scheduling
- Ins information
- Note taking & transcription

□ Medical Solutions Company

Base - Nurse

- $\frac{\text{Nurse } 80}{\text{Doctor } 15}$

- Deploy Travel Nurses

- Nursing outsourcing company

- AI project

↳ Timesheet - attstore timesheet labor.

↳ OCR solution 58% accurate

Use Case: Drug Discovery

T - ...

- Institution - AA company
- FDA only approves a handful of drugs
- They bought med research data to study time to market for drugs
- 2nd - How these compounds act on the body?
 - Allergic reactions
 - ↳ based on medical markers
 - ↳ AI can study and give warnings yes/go.

● Use Case: Remote Monitoring

- Apple can detect falls

↳ wearables record attributes 24/7

↳ Early alert

↳ notify doctor

↳ call emergency

— Dream —

— Phone

— wearable

• You have a condition

↳ enter into app

↳ predictive analysis

↳ brain condition?

↳ App gives list of specialist

↳ App gives reviews of doctors

- Doctor
- What problems are slowing it
 - PHI: personal health information
 - PII: personal identifiable Infor
 - Biggest challenge is to integrate agencies
- Bias in AI model
 - ↳ Model did not identify ethnicity affecting anesthesia.
 - Make sure data is diverse
 - rough

- Key is responsibility & diligence
- Database: Epic 3rd party support is slow
- Integrations
- regulatory hurdles
- AI ethical questions / dilemmas
 - resistance to change
 - people still trust people more

• Problems Created by AI

- Job displacement
- Humans will still be needed for supervision.
- Humans have to evolve to have qualities robots don't have.
- Over reliance on technology
 - all air ports are shutdown
 - crowd strike rushes out bank

patch.

- can

- Increased complexity

- Digital Divide

- Digital social status

↳ small hospital cannot afford latest systems

- AI systems are not affordable for everyone.

● Agriculture ●

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- What is it?
 - producing of food
 - fishing
 - farming
 - ranching

- Major use case

- Crop monitoring
 - People don't like this work
 - disease detection
- - drones
 - satellite pictures

- Precision farming

- do exactly what is needed
- plant AI apps
 - ↳ water, chemicals
 - ↳ several apps available
- automated pest control
- Yield prediction
 - will it grow this year
 - weather forecast
 - supply chain optimization
- Supply chain Optimization

- Traditional T
- high initial cost
- farmers remain poor
- regulatory hurdles
- diverse climate conditions
- Bias in AI models
- over reliance on tech
- Digital Divide persist
- AI infrastructure can have an impact on agriculture.
- Increase complexity.

